The English College in Prague:
ASGARD-XV Experiment 14 Progress Report 1

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1. Introduction:
This report shall encompass the advancements made in the past month in ASGARD-XV payload 14. The process was primarily focused on constructing the theoretical basis for the investigation, consulting external sources for model data, and preliminary preparations for the construction of the experimental module.

2. Research topic:
Extensive research was carried out into the technical components required for analysis of Single Event Upsets (further abbreviated as SEU) in SRAM chips. It was determined that SRAM chips currently available on the commercial market are excessively expensive for their relative performance, which is greatly constrained by the data transfer speed of the Arduino microcontroller. DRAM was determined to be a more suitable option for the experimental setup in regard to the price-performance ratio. However, with reference to external investigations of a similar nature, an average number of SEU, in the period of 10 minutes at 20 km of altitude, was determined to be $\approx 2.9\times10^{-5}$ to 2.9×10^{-4} using 16 gigabytes of DRAM. This suggests, by Poisson approximation, a 0.003% to 0.03% chance of seeing one or more SEUs in a time period of 10 minutes. Further options were discussed to increase the number of SEU, such as increasing the temperature of the DRAM chip, or reducing its refresh rate, however it was ultimately determined to be impossible to measure a reasonable number of SEU in the flight time window to support an accurate comparison between the level of cosmic radiation and the number of SEU.

Due to these circumstances, the **focus of the investigation was shifted** to measuring the vertical ozone concentration profile and correlating it with UVA/UVB intensity to observe how ozone absorption affects ultraviolet radiation with altitude. This research question was chosen as it repurposes a large proportion of the original experimental module and background theory, while retaining great scientific relevance. Measuring the concrete effects of the concentration of ozone in the stratosphere on the intensity of UV radiation is invaluable, with increasing fluctuations in ozone concentration consequent to terrestrial anthropogenic pollution. Regardless, there are persistent gaps in accessible in situ measurements of the stratospheric ozone layer and its interactions with ultraviolet radiation. While satellite and ground-based observations provide large-scale data, they lack the vertical resolution needed to directly correlate local ozone concentration with UV attenuation, which is highly altitude-dependent. In particular, there are relatively few student-scale balloon experiments that simultaneously measure ozone concentration and spectrally resolved UV radiation within the stratosphere.

3. Theoretical basis:

Consulting external sources, both the relative intensity of ultraviolet radiation[1] and concentration of ozone[2] were modelled against altitude:

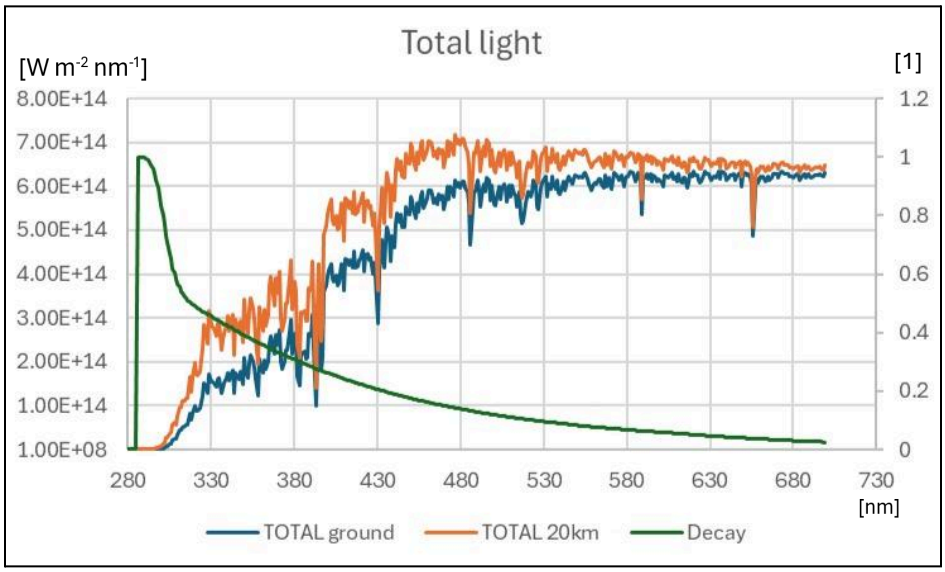

Fig. 1: A model of the decay of relative UV radiation intensity dep. on wavelength traveling from 20 km down to earth (data source: Copernicus Atmosphere Monitoring Service, 2025)

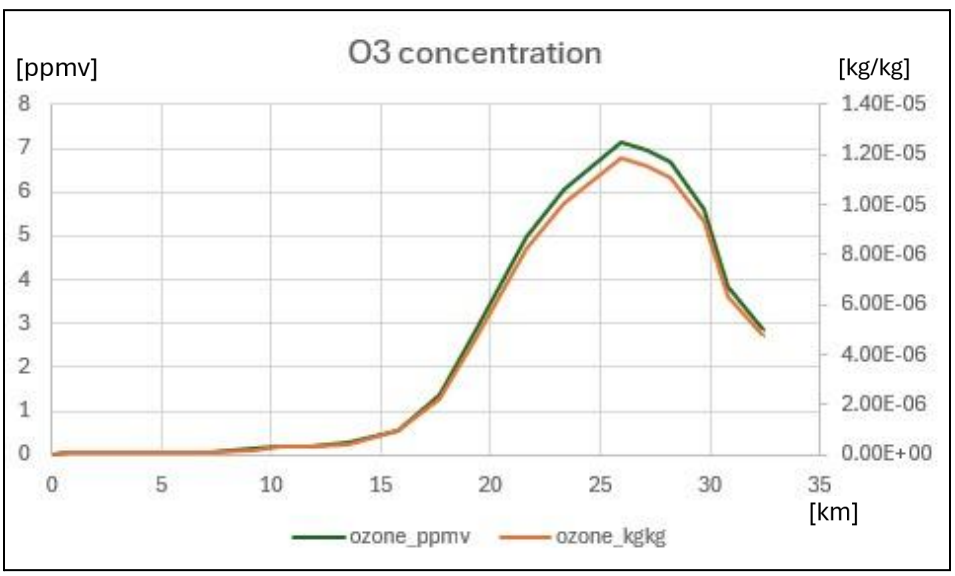

Fig. 2: A model of relative ozone concentration against altitude (University Corporation for Atmospheric Research (UCAR), n.d.)

[1]University Corporation for Atmospheric Research (UCAR) (n.d.) *Interactive TUV (Tropospheric Ultraviolet–Visible) radiation model*. Available at: http://acom.ucar.edu/Models/TUV/Interactive_TUV (Accessed: 20 January 2026).

[2]Copernicus Atmosphere Monitoring Service (2025) *CAMS global atmospheric composition forecasts.* Available at: https://ads.atmosphere.copernicus.eu/datasets/cams-global-atmospheric-composition-forecasts?tab=overview (Accessed: 12 January 2026)flightthe UV-sensitivethflighte .

The coupled calculation model was developed for this project[3]:

1. The ozone concentration prediction is downloaded from the database.
2. The contraction values are interpolated along the balloon flight path.
3. The UV spectrum is then calculated upward (toward the UV-sensitive diode) using the libRadtran software package.

The final set of model data allows for three concrete hypotheses to be formed:

- Ozone concentration will increase with altitude into the stratosphere, reach a maximum, and then level off or decrease.
- UVB intensity will show a stronger decrease in regions of higher ozone concentration than UVA.
- There will be a negative correlation between ozone concentration and UVB intensity.

This aligns with the established physics and chemistry of the stratosphere and the radiative transfer of ultraviolet radiation through ozone-rich air. Ozone is formed primarily in the stratosphere through photochemical reactions initiated by solar ultraviolet photons, where molecular oxygen is dissociated and subsequently recombines to form ozone. As altitude increases from the troposphere into the stratosphere, ozone concentration rises because UV radiation becomes sufficiently intense to drive these reactions while air density remains high enough to allow stabilizing molecular collisions. At higher altitudes, decreasing air density limits ozone production despite stronger UV flux, producing a characteristic ozone maximum rather than a continuous increase. At the same time, ozone's absorption of ultraviolet radiation is strongly wavelength-dependent. Ozone has a much larger absorption cross-section in the UVB band than in the UVA band, meaning UVB is preferentially attenuated as radiation passes through ozone-rich layers. As a result, UVB intensity is expected to decrease more sharply with increasing ozone concentration than UVA, which is affected primarily by scattering rather than absorption. This relationship follows directly from the Beer–Lambert law, where the transmitted UV intensity decreases exponentially with increasing optical depth. Because optical depth in the UVB range is dominated by ozone absorption, regions of higher ozone concentration correspond to lower measured UVB intensity. Together, these physical principles predict a peak in ozone concentration within the stratosphere, a pronounced attenuation of UVB in that region, and a clear inverse correlation between ozone concentration and UVB intensity as altitude varies.

4. Experimental design:

The experimental module has two sections: an external one mounted outside the gondola and an internal one mounted inside. Both sections are interconnected by wires leading from the gondola outside.

The first component of the external section is an ozone sensor, which will measure the concentration of ozone in the surrounding atmosphere. The second component is a UV sensor module, which will measure the intensity of the incoming ultraviolet radiation. The third component is a joint air pressure, temperature, and humidity sensor, which records these values and allows us to separate the effects of these factors from the isolated effects of ozone concentration during interpretation. These modules are attached to a mini breadboard.

The internal section is made up of components enclosed within a 3D printed, thermally insulated casing to protect against temperature extremes. The first component of the internal section is an Arduino

[3] https://github.com/jdobes-cz/stratospheric-ozone-uv-profile/tree/main

microcontroller, which controls and gathers readings from all other components of the experiment. The second component is an RTC module, which timestamps all recordings from the experiment with the current time, preventing discontinuity if the Arduino resets. The third component is an SD card module or memory module, which stores the readings of all other components of the experiment for later interpretation.

5. Preliminary parts list:

Sensor/Board type	Voltage	Pins	Operating range	Mass + power consumption	URL of the product	URL of data sheet
Microcontroller: Arduino Pro Mini (ATmega328PB, 3.3V/8MHz)	Logic/supply: 3.3V; board input: 5–12V (RAW) or 3.3V (VCC) (laskakit.cz)	14× digital I/O (6 PWM), 8× analog; 1× SPI, 1× I2C, 1× USART (laskakit.cz)	ATmega328PB operating temperature (datasheet): −55 to +125 °C (Microchip)	Weight: 11 g (laskakit.cz); active supply current shown as curves (e.g., 8 MHz vs VCC) (Microchip)	https://www.laskakit.cz/en/arduino-pro-mini--atmega328p-tqfp-3-3v-8mhz/	https://ww1.microchip.com/downloads/aemDocuments/documents/MCU08/ProductDocuments/DataSheets/40001906C.pdf
Ozone sensor: DFRobot SEN0321 Gravity I2C O_3 (0–10 ppm)	3.3–5.5 V (Farnell)	I2C: SDA, SCL (+VCC, GND; "Gravity-4Pin") (Farnell)	O_3: 0–10 ppm; **−20** to +50 °C; **15**–95% RH **ambient pressure** (Elektronikos parduotuvė | Anodas.lt)	Weight: 22 g (Elektronikos parduotuvė | Anodas.lt); power/current: not specified in seller specs (described as "low power consumption") (Arduino Official Store)	https://cz.farnell.com/dfrobot/sen0321/gravity-i2c-ozone-sensor/dp/3769917	https://robu.in/wp-content/uploads/2021/12/3c1da72292464bac7c2761d932e72b1c.zh-CN.en_.pdf

UV + ambient light: LaskaKit LTR390 UV sensor module	Module supply: 3.3–5 V (laskakit.cz); IC operating voltage: 1.7–3.6 V	I2C: SDA, SCL (+optional INT, plus VCC/GND)	IC operating temperature: **−40** to +85 °C	Weight: 2 g (laskakit.cz); module consumption listed: 2.73 mA (laskakit.cz); IC current: ~110 µA (ALS) / 100 µA (UVS), standby ~1 µA	https://www.laskakit.cz/en/laskakit-ltr390-uv-senzor	https://eu.mouser.com/datasheet/3/281/1/LTR_390UV_Final_DS_V1_1.pdf
Temperature (thermocouple ADC): MAX31856 SPI module	3.3 V / 5 V (module) (laskakit.cz); IC supply: 3.0–3.6 V	SPI: SCK, SDI(MOSI), SDO(MISO), CS (+DRDY/FAULT optional)	IC operating temperature: −55 to +125 °C; thermocouple temperature range depends on type (K/J/N/R/S/T/B/E supported)	Weight: 3 g; module consumption listed: 2.2 mA (laskakit.cz); IC current: standby 5.25–10 µA, active conversion 1.2–2 mA	https://www.laskakit.cz/en/univerzalni-modul-prevodniku-pro-termoclanky--max31856--spi/	https://www.analog.com/media/en/technical-documentation/data-sheets/max31856.pdf
Pressure: MS5611-01BA03 barometric pressure breakout	Sensor VDD: 1.8–3.6 V (TE Connectivity)	I2C or SPI (TE Connectivity)	**10** to 1200 mbar; **−40** to +85 °C (TE Connectivity)	Sensor package: QFN 5×3×1 mm (TE Connectivity); sensor unit weight ~0.034 g (Mouser Electronics) (breakout-board weight not specified); current: "low power	https://techfun.cz/produkt/senzor-atmosferickeho-tlaku-ms5611-01ba03	https://www.te.com/commerce/DocumentDelivery/DDEController?Action=showdoc&DocId=Data+Sheet%7FMS5611-01BA03%7FB3%7Fpdf%7FEnglish%7FENG_DS_MS5611-01BA03_

				1 µA (standby <0.15 µA)", peak during conversion 1.4 mA (TE Connectivity)		B3.pdf%7FCAT-BLPS0036
Memory: AT24C256 I2C EEPROM module	IC variants support 1.8–3.6 V and 2.7–5.5 V (overall: 1.8–5.5 V family) (Microchip)	I2C: SDA, SCL (+A0/A1 address, WP, VCC, GND) (Microchip)	Recommended operating ambient: −40 to +85 °C; extended −40 to +125 °C for "B" process at higher VCC (Microchip)	Module weight: 0.001 kg (1 g) (laskakit.cz); IC current (typ @ 5V, 400 kHz): read 1–2 mA, write 2–3 mA; standby as low as 0.2 µA (1.8V option) (Microchip)	https://www.laskakit.cz/eeprom-modul--at24c02--i2c/	https://ww1.microchip.com/downloads/en/DeviceDoc/doc0670.pdf
Real-time clock: DS3231 (RTC) module	Module supply: 3.3 V (laskakit.cz); IC VCC range: 2.3–5.5 V	I2C: SDA, SCL (+SQW/INT, 32 kHz out, VBAT depending on module)	IC operating temp (DS3231SN): −40 to +85 °C	Module weight: 0.004 kg (4 g) (laskakit.cz); IC timekeeping current (typ): 110 µA at 3.63 V; active supply current (typ): 200 µA	https://www.laskakit.cz/laskakit-ds3231-orig--rtc-hodiny-realneho-casu/	https://www.analog.com/media/en/technical-documentation/data-sheets/ds3231.pdf

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